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EDITED BY
Xiaofei Wang,
Beihang University, China

REVIEWED BY
Runze Li,
Massachusetts Institute of Technology, United States
Lin Liu,
Louisiana State University, United States

*CORRESPONDENCE
Liya Qiao,
dr_qiaoliya@163.com
Xiuqing Qian,
qianxq@ccmu.edu.cn

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Biomechanical responses and scleral reinforcement across staphyloma subtypes in pathologic myopia: a finite element analysis

Yanqing Wang[1], Xiaoxia Li[2], Jifeng Ren[1,3,4], Liya Qiao[5]* and Xiuqing Qian[1,3,4]*

[1]School of Biomedical Engineering, Capital Medical University, Beijing, China, [2]Department of Ophthalmology, Beijing Shijitan Hospital, Capital Medical University, Beijing, China, [3]Beijing Key Laboratory of Intelligent Diagnosis Technology and Equipment for Optic Nerve-Related Eye Diseases, Beijing, China, [4]Beijing Key Laboratory of Clinical Engineering Solutions for Mental Health, Beijing, China, [5]Beijing Tongren Eye Center, Beijing Tongren Hospital, Capital Medical University, Beijing, China

Objective: Posterior scleral staphyloma (PSS) is a common complication of pathologic myopia and may lead to deformation of posterior pole structures, including macula and lamina cribrosa (LC). Different types of PSS exhibit distinct structural features and corresponding patterns of fundus damage. Prior studies have defined PSS morphological classification and its association with fundus lesions, yet the biomechanical mechanisms of subtype-specific damage and quantitative evaluation of posterior scleral reinforcement (PSR) remain unaddressed. Therefore, this study explored posterior pole biomechanical responses in PSS and PSR efficacy to guide clinical management.

Methods: Whole-eye finite element models were established incorporating five Curtin-classified PSS subtypes and three axial lengths. Key posterior pole biomechanical parameters related to lamina cribrosa displacement and macular strain were extracted. PSR was then simulated using the reinforcing strip with varied elastic moduli to compare pre- and post-operative mechanical responses.

Results: PSS subtype and staphyloma depth-radius morphological combinations influenced biomechanical responses of posterior pole structures. Type I (wide macular) PSS demonstrated the largest LC displacements, with mean LC_X_DistMean of 0.0215 ± 0.0133 mm and LC_Y_DistMean of 0.0188 ± 0.0138 mm. HSR_RM_0.04 was lower in the PSR models (from 19.45% to 3.58%), indicating reduced macular high-strain exposure. Higher strip modulus further decreased HSR_RM_0.04 across all five PSS subtypes.

Conclusion: PSS subtype-specific geometries showed the largest effect within the tested parameter ranges, while the depth-radius configuration accounted for subtype-specific variations in macular strain patterns. PSR simulations indicated reduced IOP-driven HSR_RM_0.04, enhanced by higher strip modulus with the strip configuration used in this study. These findings support morphology-tailored, *in silico* comparisons within static idealized models and warrant prospective clinical validation.

KEYWORDS
finite element analysis, pathologic myopia, posterior scleral reinforcement, posterior scleral staphyloma, staphyloma morphology

1 Introduction

Pathologic myopia (PM) is a leading cause of irreversible blindness worldwide. Epidemiological data showed that the global number of people with myopia has reached 1.41 billion, including approximately 160 million with high myopia (2.7% of the global population). It is projected that by 2050, the number of high myopia cases will reach to 940 million (9.8%) (Holden et al., 2016). In 2015 alone, myopic macular degeneration (MMD) caused visual impairment in 10 million people globally, of whom 3.3 million were blind (Fricke et al., 2018). PM is driven by axial elongation triggering blinding complications, such as myopic maculopathy (including macular retinoschisis, choroidal neovascularization, and macular hole), glaucoma, and rhegmatogenous retinal detachment (Ohno-Matsui K. et al., 2016; 2019).

Posterior scleral staphyloma (PSS) is a typical sign of PM, characterized by abnormal outpouching of posterior ocular wall due to scleral thinning and ectasia (Spaide, 2014; Shinohara et al., 2017), which could lead to deformation of the macula and the lamina cribrosa (LC), leading to macula and optic nerve damage (Ohno-Matsui K. et al., 2016; 2019). Clinical studies have primarily focused on the morphological definition and classification of PSS through correlative analyses of its specific complications (Curtin, 1977; Steidl and Pruett, 1997; Spaide, 2014), while research on its underlying biomechanical mechanisms has been rarely reported. Different types of PSS exhibit distinct structural features and corresponding patterns of fundus damage. Currently, the main clinical intervention for PSS is posterior scleral reinforcement (PSR), but the postoperative recovery outcomes of patients remain unstable and controversial (Ma et al., 2022; Miao et al., 2019). Therefore, it is important to investigate the biomechanical esponse of posterior pole structures in eyes with PSS.

PSS is associated with scleral thinning, choroidal atrophy, and Bruch's membrane defects (Jonas et al., 2020; Ehongo, 2023). Advanced 3D magnetic resonance imaging (MRI) has quantified significant scleral and choroidal thinning concomitant with global eyeball deformation in myopic eyes (Luo et al., 2023), while ultra-widefield optical coherence tomography angiography (OCTA) studies further identify reduced retrobulbar blood flow velocity as a critical hemodynamic driver in PSS pathogenesis (Liu F. et al., 2025). This synergy of structural deformation and microcirculatory impairment not only exerts mechanical traction on retina and LC but also precipitates irreversible neural tissue damage, characterized by significant thinning of retinal nerve fiber layer and substantial loss of retinal ganglion cells (RGCs) (Delaunay et al., 2025). Collectively, these insults culminate in vision-threatening complications, including myopic maculopathy and optic neuropathy, which correlate closely with severe visual acuity decline (Flores-Moreno et al., 2024).

Based on the topography of outpouching, Curtin (Curtin, 1977) classified simple PSS into five distinct subtypes: Type I (wide macular), Type II (narrow macular), Type III (optic nerve head/peripapillary), Type IV (nasal), and Type V (inferior). The wide macular type was the most prevalent (Curtin, 1977; Tanaka et al., 2019). Accumulating evidence suggest significant heterogeneity among these subtypes in their biomechanical impact and resultant fundus damage patterns (Ruiz-Medrano et al., 2024). Notably, different PSS subtypes exhibit distinct clinical prognoses. In Curtin's original study (Curtin, 1977), 34.5% of eyes with Type I staphyloma had visual acuity of 20/200 or worse, and 19% met the legal blindness criteria. Wide macular posterior staphyloma (WMPS) was subclassified into primary and compound forms (corresponding to Curtin types I and VI–X, respectively). Compared to the primary form, compound WMPS demonstrated a higher prevalence of atrophic and tractional complications and poorer visual outcomes (Yoon et al., 2024). In contrast, Type III, centered on the optic nerve head (ONH), imposes direct mechanical stress on the optic nerve, leading to distinct patterns of optic neuropathy and visual field defects (Luo et al., 2023; Ruiz-Medrano et al., 2024). Recent evidence links posterior scleral topography to ONH biomechanics, with high myopia inducing a significantly more negative LC tilt despite no significant difference in LC depth (Han et al., 2024a). Furthermore, topographic parameters (Dvar, CDmax) obtained by MRI correlated strongly with myopic traction maculopathy severity (Han et al., 2024b).

The finite element method (FEM) has been widely applied in ophthalmic biomechanical research. Researchers have established different models to quantify the retinal deformation associated with intraocular pressure (IOP) elevation, as well as LC strain correlated with both scleral thickness and stiffness (Sigal et al., 2004; Wei J. et al., 2022; Islam et al., 2024). To explore the influence of ocular movement on ocular deformation in patients with high myopia, Liu et al. (Liu T. et al., 2025a; 2025b) constructed a quantitative analysis framework for ONH deformation. Based on the finite element model of healthy human eyeball, Braeu (Braeu et al., 2024) simulated growth and structural remodeling of model over time to investigate the mechanism of PSS formation. These results indicated that the non-uniform distribution of scleral stiffness and adaptive changes in collagen fiber orientation are key factors leading to local weakening and expansion of the posterior pole of the eyeball.

The highly individualized morphology of PSS critically governs the ocular mechanical response. Theoretically, the distinct subtypes impose varied stress and strain distributions depending on their locations, depths, and extents. Current understanding of the underlying biomechanics is still nascent, highlighting an urgent need for refined modeling to predict these mechanical responses under physiological loading. Given the established link between PSS morphology and fundus damage, investigating these biomechanical responses is imperative for deciphering the pathogenesis of vision loss and optimizing clinical interventions. This study aims to use FEM to establish whole-eye geometric models that incorporate key structures including retina, choroid, sclera, optic nerve, and LC. The model is designed to explore the mechanical responses of the macular center and ONH region under varying conditions, including different PSS subtypes, diverse staphyloma morphological features, and different IOP levels.

2 Materials and methods

2.1 Whole-eye geometric model for PSS

Finite element analysis was performed using ANSYS 2024 R^2 (ANSYS, Inc., Canonsburg, PA, United States of America).

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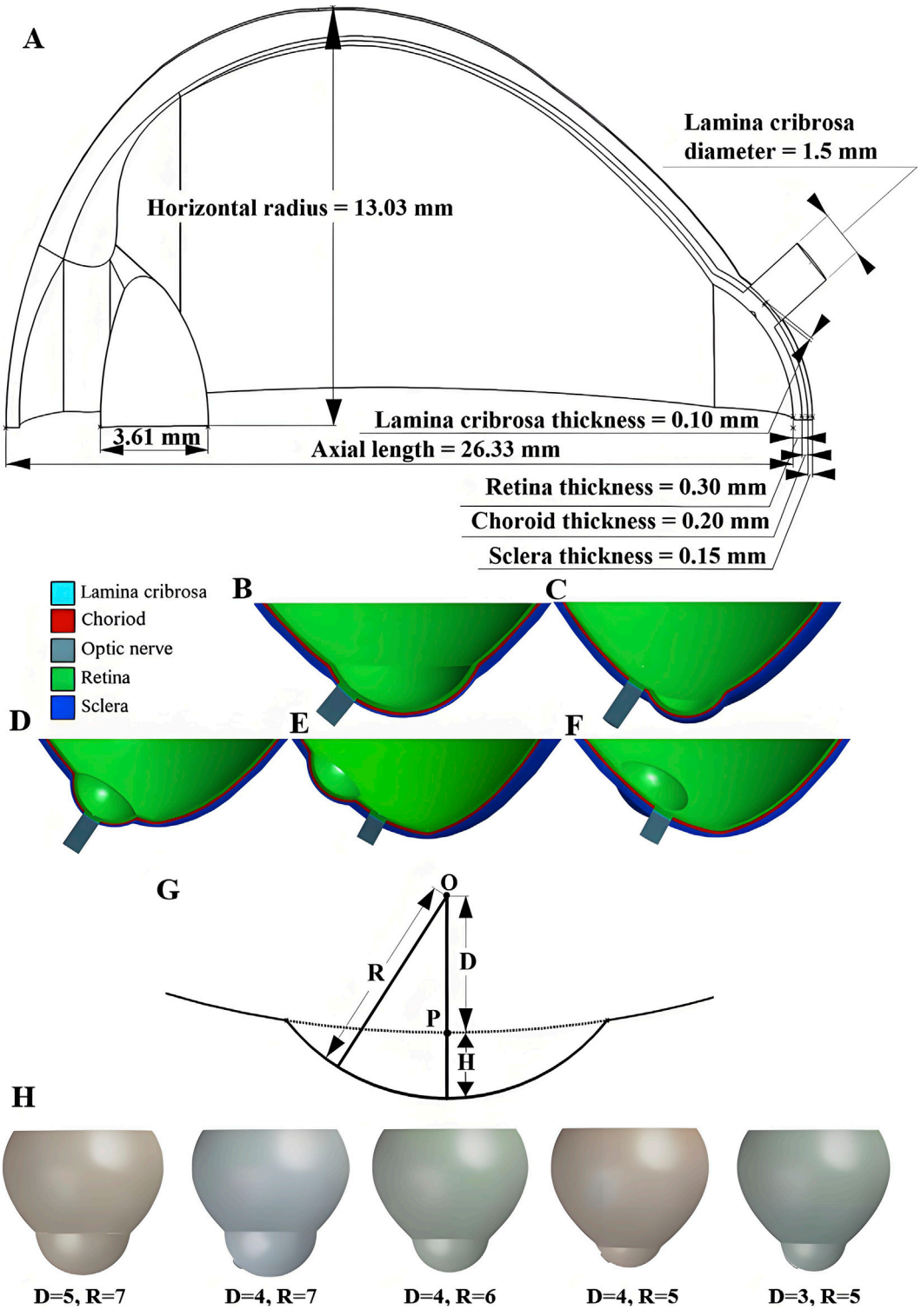


FIGURE 1
Computational modeling framework of the whole-eye finite element model with posterior staphyloma (PSS). **(A)** Anatomical overview of the whole-eye model (B ~ F): Schematic representation of the five primary PSS subtypes. **(B)** Type I (wide macular); **(C)** Type II (narrow macular); **(D)** Type III (optic nerve head); **(E)** Type IV (nasal); **(F)** Type V (inferior). (G ~ H) Schematic diagram of the geometric definition of PSS morphological parameters **(G)** Geometric definition of PSS morphological parameters. P Reference point, located on the outer retinal surface, used to determine the position of the staphyloma; **(D)** Depth, the distance extended vertically inward from point P; O: Origin, the coordinate origin determined by extending inward from point P by depth D; R: Radius, the spherical radius used to fit the depressed staphyloma; H (Height): The protruding height of the staphyloma. **(H)** Schematic diagram of PSS morphological parameters.

TABLE 1 Tissue thicknesses (mm) in different regions.

Region	Sclera	Choroid	Retina
Within staphyloma	0.15	0.20	0.30
Outside staphyloma	0.50	0.25	0.30
Optic nerve head (ONH)	0.15	0.15	0.25
Fovea	0.20	0.20	0.30

Computational tasks were executed on a workstation configured with the following hardware and software components, including an AMD Ryzen 7 5700X 8-core processor, 48 GB DDR4 RAM, and a Windows 11 operating system. The whole-eye model (Figure 1A) was designed based on the study by Liu X et al. (Liu X. et al., 2013), incorporating the cornea, sclera, ciliary body, choroid, retina, lamina cribrosa, optic nerve, lens, and zonules. The transverse diameter was set as 25 mm for all models. Based on the reported thickness data (Cui et al., 2021; Wei et al., 2013; Hayashi et al., 2013), the tissue thicknesses of the sclera, choroid, and retina were defined for four anatomical regions (Table 1).

Eyeballs with PSS were classified into five subtypes (Figures 1B-F) based on the Curtin (Curtin, 1977) classification, and characterized by four parameters quantifying their morphological features and dimensions: P (Position), O (Origin), D (Depth), and R (Radius of Staphyloma). After reference point P on the outer retinal surface was selected, a perpendicular inward extension of depth D defined the coordinate origin O (Figure 1G). The surface at a distance of R from origin O represents the outer surface of retina, shaping the bulged staphyloma. Parameter P determines the location of the staphyloma, while D and R define its morphological characteristics. Figure 1H presents the typical morphologies for different D and R values.

According to reference (Chen Y. et al., 2023), the LC thickness in patients with high myopia combined with primary open-angle glaucoma (POAG) is reported to be 107.76 ± 9.86 μm, whereas it is 137.07 ± 18.51 μm in patients with high myopia alone. Jonas et al. (2004) reported that the central LC thickness in highly myopic eyes with glaucoma is significantly reduced, with an average thickness of approximately 77.7 μm, compared to 206.7 μm in highly myopic eyes without glaucoma. Therefore, the LC thickness was uniformly set to 100 μm in all models of our study.

2.2 Material parameters, mesh and boundary conditions

The material properties of ocular tissues included in the finite element model are presented in Table 2. To better capture the large shape changes in the macular area and the ONH region, we applied a finer mesh to these two focal areas of the study. Globally, element size of 1 mm was set for most regions, while the ONH and macula region were refined to 0.2 mm. The fovea and ONH were further refined to 0.06–0.08 mm. To avoid stress concentration, the size ratio between adjacent transition zones is limited under 2:1. Mesh independence was verified (Figure 2) and approximately 35,000 elements were used for all experimental models. Fixed boundary condition was applied at the corneoscleral junction, and IOP of 10 or 21 mmHg was applied to the retinal inner surface. A frictionless contact is established between the sclera and the lateral side of the optic nerve, allowing unrestricted tangential relative motion and normal separation. Other tissue junctions are set as bonded connections, restricting both normal and tangential relative motion.

2.3 Model classification

Model differences were quantified through 7 specific parameters (Table 3). The classification of the five PSS subtypes was initially proposed by Curtin (Curtin, 1977), in our study were distinguished by the position of point P and the coverage area of the staphyloma.

Type I (Wide Macular): P at macula center, covers both ONH and macula.

Type II (Narrow Macular): P at macula center, covers macula (excludes ONH).

Type III (Optic Nerve Head): P at ONH center, covers ONH (excludes macula).

Type IV (Nasal): P at ONH nasal side, covers neither ONH nor macula.

Type V (Inferior): P inferior to ONH, covers neither ONH nor macula.

We selected AL of 26 mm, 29 mm, and 36 mm to represent the mild, moderate, and extreme axial elongation, respectively, for the simulation experiments (Curtin and Karlin, 1971; Curtin, 1977; Knaus et al., 2021; Luo et al., 2025). Subtype-specific D and R combinations followed Table 3 (Figure 1H). IOP was 10 or 21 mmHg.

Given that PSS morphology directly influences posterior pole mechanical state, we next evaluated whether surgical treatment could prevent these mechanical risks. The PSR procedure was simulated using the single wide strip method (Xue et al., 2014; Ward et al., 2009) with different elastic modulus. In PSR-treated models (Figure 3), each incorporates a 5-mm-wide, 1-mm-thick reinforcement strip applied symmetrically at the macular center. For surgical modeling parameters, the elastic modulus of reinforcement strips (3–50 MPa) covered the spectrum from native sclera to clinical grafts (40–70 MPa) (Ma et al., 2022). This framework anchors our findings on PSS subtype grouping, Depth–Radius synergy in regulating macular strain, and the dose-dependent effect of strip rigidity.

2.4 Statistical analysis

We assessed the impact of geometric parameter changes on biomechanical responses under a non-parametric statistical framework by testing the intergroup differences for statistical significance. The Kruskal–Wallis test (Kruskal and Wallis, 1952) avoids distributional assumptions through rank transformation and is suitable for continuous or ordinal data. However, its test statistic H only provides significance judgments and does not quantify effect size. To address this, effect size measures are introduced to complement the P-value, directly reflecting the magnitude of intergroup differences without being influenced by sample size fluctuations, thereby enhancing result

TABLE 2 Material properties of ocular tissues.

Tissue	Constitutive Model	Material properties		References
		Young's modulus (MPa)	Poisson's ratio	
Ciliary Body	Isotropic Elasticity	0.125	0.49	(Knaus et al., 2021)
Zonules		5.15	0.49	(Knaus et al., 2021)
Lamina Cribrosa		0.3	0.49	(Zhang et al., 2015)
Optic Nerve		0.03	0.49	(Miller, 1999)
Choroid		0.6	0.49	(Friberg and Lace, 1988)
Retina		0.03	0.49	(Miller, 1999)
Sclera		3	0.49	(Sigal et al., 2005)
Cornea	Neo-Hookean	Initial Shear Modulus = 1 MPa Incompressibility Parameter = 4E-10 Pa^{-1}		(Wang H. et al., 1996)
Lens		Initial Shear Modulus = 0.2 MPa Incompressibility Parameter = 4E-10 Pa^{-1}		(Krag and Andreassen, 2003)

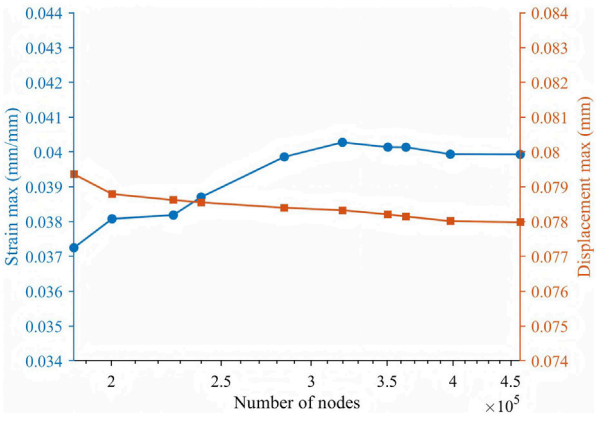


FIGURE 2
Mesh convergence. Convergence of LC maximum equivalent strain (left axis, mm/mm) and LC maximum displacement (right axis, mm) with increasing mesh node count. The node number is plotted on a logarithmic scale.

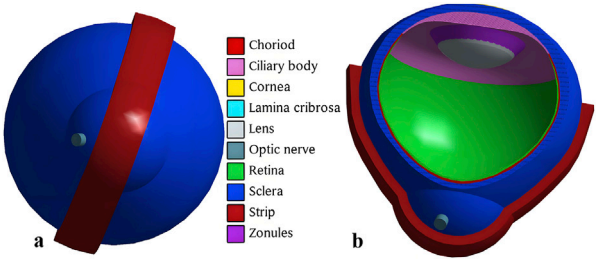


FIGURE 3
Schematic diagram of the postoperative model for posterior scleral reinforcement (PSR). **(a)** Posterior view of the PSS surgical plan in the established finite element model; **(b)** Temporal view.

TABLE 3 Model parameter ranges.

Parameter	Value range
Staphyloma Type (Type)	Type I–V
Axial length (mm)	26, 29, 36
D, R — Type I (mm)	D = 3, 4, 5; R = 5, 6, 7
D, R — Types II–V (mm)	D = 1.75, 2, 2.25; R = 3, 3.25, 3.5
IOP (Intraocular Pressure, mmHg)	10, 21
Surgery status	No/Yes
E (Elastic modulus of strip, MPa)	3, 4, 5, 10, 20, 30, 40, 50

D/R ranges differ by subtype and are not fully crossed. PSR models use an axial length of 26 mm only. Type I uses D = 4 mm and R = 6 mm. Types II -V use D = 2 mm and R = 3.25 mm. The eight strip moduli are listed above.

reproducibility and clinical interpretability (Kruskal and Wallis, 1952; Fiel Peres, 2026).

To quantify the explanatory power of geometric parameters (PSS subtype, axial length, depth D, radius R, and surgical status) on the variability of mechanical responses, the effect size metric ε^2 specific to the Kruskal–Wallis test is adopted. Its calculation formula is given in Equation 1:

$$\varepsilon^2 = \frac{H-(k-1)}{N-k} \quad (1)$$

where H is the test statistic of the Kruskal–Wallis test, reflecting the deviation of intergroup dispersion of mechanical results relative to intragroup dispersion; k is the number of groups for the geometric parameter, representing the studied geometric feature differences. H–(k–1) denotes the intergroup variability adjusted for degrees of freedom, and N–k is the total degrees of freedom. The theoretical range of ε^2 is 0–1 and values closer to 1 indicate a higher proportion of variability explained by the geometric parameters. Generally, $\varepsilon^2 \geq 0.14$ can be considered a large effect, suggesting that the grouping factor has strong influence (Fiel Peres, 2026).

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TABLE 4 Number of models by PSS subtype

PSS subtype	Non-surgical	Surgical	Total
Type I	24	8	32
Type II	27	8	35
Type III	25	8	33
Type IV	27	8	35
Type V	27	8	35
Total	130	40	170

Identical distribution for both IOP, groups (10 and 21 mmHg), each with 170 samples (Total: 340). The 8 surgical models per subtype correspond to 8 discrete scleral moduli (3–50 MPa). Variation in Non-surgical counts reflected the exclusion of non-convergent models where D/R ratios approximated 1. Type I lacks D = 5 mm with R = 5 mm; Type III, lacks axial length 29 mm with D = 2.25 mm and R = 3 mm, and axial length 36 mm with D = 1.75 mm and R = 3.25 mm.

3 Results

A total of 340 static whole-eye models were computed (Table 4), corresponding to two IOP loading conditions (10 mmHg and 21 mmHg) applied to the same set of eye morphologies. The differences in mechanical responses (i.e., displacement, strain, and stress) between the two IOP conditions were then used for subsequent statistical analysis. Data extracted from ANSYS Mechanical were subsequently post-processed using MATLAB R2021b (MathWorks, Natick, MA, United States of America).

3.1 Determination of key biomechanical parameters

Geometric variations primarily affect maximum strain and stress. These parameters are more susceptible to segmentation boundary uncertainty, mesh discretization errors, and noise (Li et al., 2026). We prioritized average and region-specific metrics. Previous studies have identified IOP-induced LC displacement and scleral canal expansion as critical parameters for assessing ONH biomechanical response (Sigal et al., 2011). And further research showed that scleral geometry variations significantly influence LC strain and canal deformation, underscoring the need to capture region-specific rather than global metrics (Norman et al., 2011). Anisotropic scleral and pial mechanics, along with optic nerve sheath traction, have also been shown to produce distinct ONH stress–strain patterns, validating the use of region-specific displacement and strain measures to characterize biomechanical insult (Karimi et al., 2022; Shin et al., 2017).

Therefore, we selected four key parameters to quantitatively characterize the biomechanical response of the ONH and macular regions, using regional displacement for the LC and regional strain for the macula.

(1) Mean displacement in the radial direction of the lamina cribrosa (LC_X_DistMean).
(2) Mean displacement in the circumferential direction of the lamina cribrosa (LC_Y_DistMean).
(3) High-strain ratio in macula region (defined by strain >4%) (HSR_RM_0.04).
(4) Internal strain ratio in the retinal macula (ISR_RM_0.25).

A cylindrical coordinate system was constructed on the upper surface of the LC, where X denotes the radial direction and Y the circumferential direction. The mean displacement in the X and Y directions reflects the overall displacement magnitude of the LC region in the radial and circumferential directions, respectively. We extracted X and Y displacement components from LC nodes and took absolute values to remove directional dependence. Invalid points were the LC junction ring with sclera, choroid, and retina, where displacement fluctuated sharply; these were excluded before computing the mean absolute displacement. The LC thickness was 0.1 mm and the circular diameter 1.5 mm; with mesh size <0.08 mm, elements more than 0.7 mm from the center were excluded, amounting to about 10%.

The high-strain ratio HSR_RM_0.04 in macular region is used to quantify the frequency of high-strain occurrences. Initially, the centroid coordinates of the strain distribution in the macular area were determined. The Euclidean distance from each strain element to the centroid was subsequently measured. With the macular center set as the origin and an analysis radius of 0.75 mm, strain data within this region were collected. The 0.75 mm radius follows the anatomical fovea, whose diameter from rim to rim is about 1.5 mm (Polyak, 1941; Kolb et al., 2020). We did not use the maximum value of strain, because the maximum value could be local extrema caused by mesh distortion and may not be physically meaningful. Then a high-strain threshold of 4% was applied to identify all strain elements exceeding this value. HSR_RM_0.04 was defined as the percentage of elements located within 0.75 mm of the macular center with an equivalent strain ≥4%, relative to the total number of elements in that region.

The internal strain ratio ISR_RM_0.25 in the retinal macula quantifies the strain contribution proportion of the core macular region. The analysis area was delineated by an inner circle radius of 0.25 mm and an outer circle radius of 0.75 mm. The sum of strain values within the inner circle (inner circle strain contribution) and the sum of strain values across the entire macular analysis area (total strain contribution) were computed. ISR_RM_0.25 was derived as (inner circle strain contribution/total strain contribution) × 100%.

Overall, LC_X_DistMean and LC_Y_DistMean were 12.21 ± 9.26μm and 9.88 ± 9.50μm, respectively. Peak stresses in the ONH and fovea regions were 3.07 ± 1.29 kPa and 3.22 ± 2.10 kPa, respectively, while maximal principal strains were 11.08% ± 4.80% and 10.84% ± 4.48%. The maximal equivalent LC strain was 7.05% ± 3.74%, while maximal principal strain was 5.37% ± 2.78%.

The maximum strain was systematically observed at the retina-LC interface (Figure 4). Pearson correlation results for continuous variables (Figure 5) indicate that LC_X_DistMean and LC_Y_DistMean exhibit high correlations with the maximum LC equivalent strain and the maximum equivalent strain of the ONH. The maximum optic-nerve strain was strongly correlated with the maximum equivalent strain of the lamina cribrosa (r = 0.737). Similarly, HSR_RM_0.04 and ISR_RM_0.25 show high correlations with the maximum equivalent strain of the macular retina.

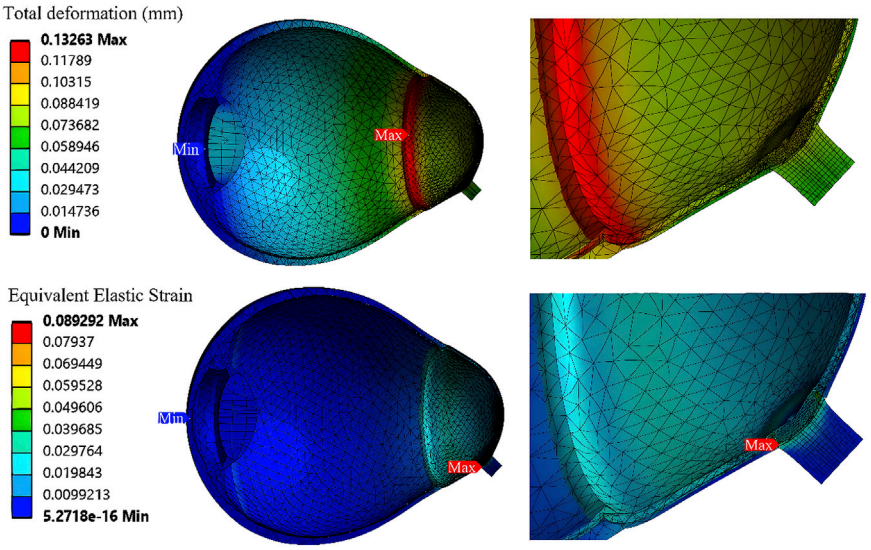


FIGURE 4
Contour plot of deformation and strain.

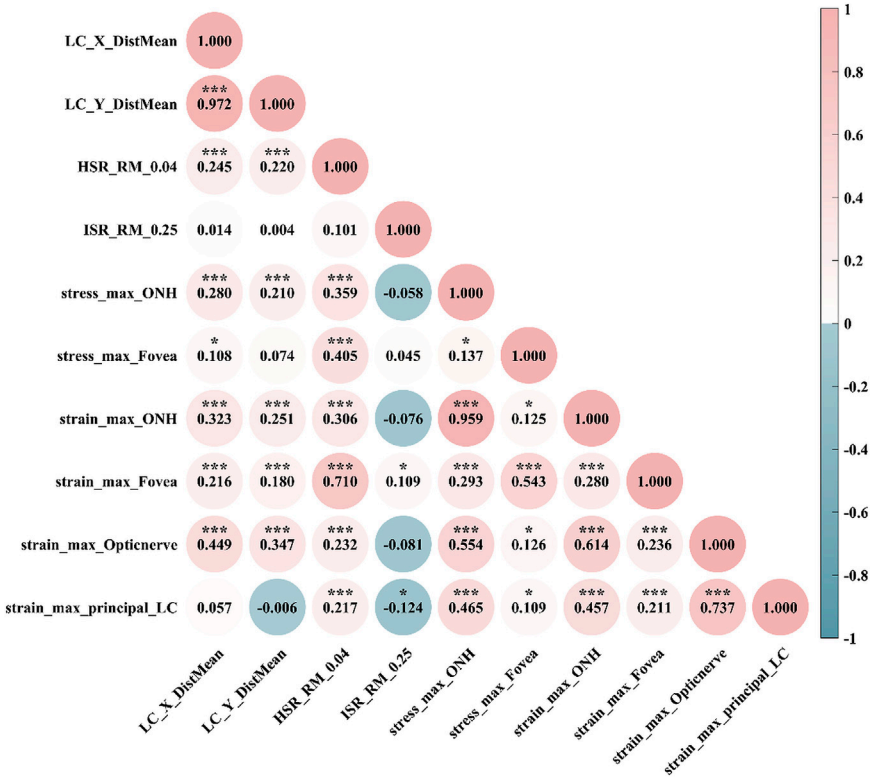


FIGURE 5
Pearson correlation heatmap of continuous variables in finite element model.

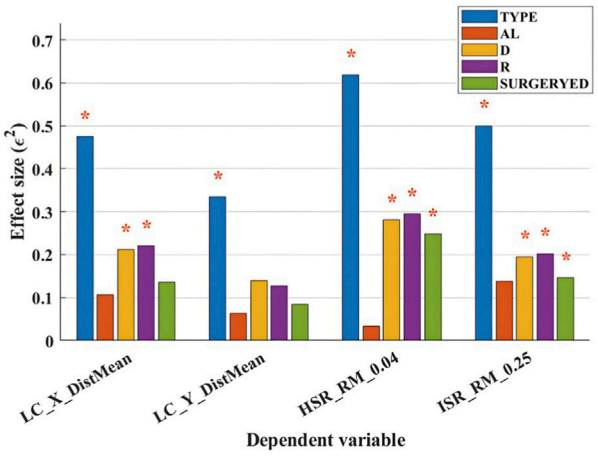


FIGURE 6
Effect sizes (ε^2) of geometric parameters on biomechanical parameters. X-axis, four outcomes; legend, five geometric factors; bar height, ε^2. Asterisks (*), $p < 0.001$, significant H, and $\varepsilon^2 > 0.14$.

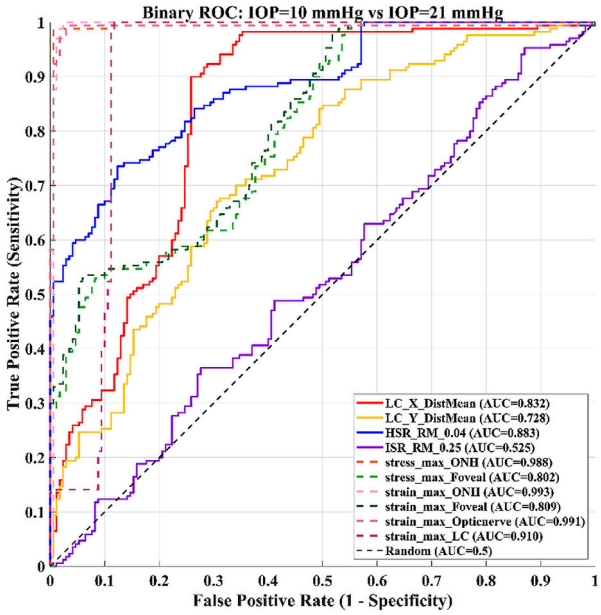


FIGURE 7
ROC curves for discriminating an IOP of 10 mmHg from 21 mmHg.

3.2 PSS subtype-specific geometries showed the largest effect within the tested parameter ranges

The explanatory power of geometric parameters (PSS subtype, axial length, depth D, radius R, and surgical status) on the variability of biomechanical responses was evaluated using the Kruskal–Wallis test. The analysis revealed that the effect size ε^2 for PSS subtype ranged from 0.335 to 0.619 (Figure 6). In contrast, the effect size ε^2 for axial length (AL) ranged under 0.136 and surgical status showed only under 0.144. And depth D and radius R exhibited moderate-to-large effects on select parameters (ε^2 ranging from 0.127 to 0.294). Kruskal–Wallis test results suggest that PSS subtype-specific geometries showed the largest effect within the tested parameter ranges.

Four key parameters differed significantly among the five PSS subtypes (Kruskal–Wallis, $p < 0.001$ for each). For LC displacement, Type I showed the largest mean values in both directions (LC_X_DistMean, 0.0215 ± 0.0133 mm; LC_Y_DistMean, 0.0188 ± 0.0138 mm), whereas Type V exhibited the smallest displacements (LC_X_DistMean, 0.0078 ± 0.0035 mm; LC_Y_DistMean, 0.0057 ± 0.0037 mm), with Type II and Type IV also relatively low. HSR_RM_0.04 was highest in Type III (24.22 ± 30.67) and lowest in Type I (0.97 ± 0.87), with Type V showing the second-highest mean (11.22 ± 11.63). For ISR_RM_0.25, Type III (24.22% ± 30.67%)" , "Type I (0.97% ± 0.87%)" , "Type V (11.22% ± 11.63%)" , "Type III had the highest mean (12.26% ± 2.39%)" , and "Type I the lowest (10.64% ± 1.14%), with the remaining subtypes showing intermediate values.

3.3 Discriminative value of key parameters

To identify indicators capable of effectively discriminating between subtypes, we employed the One-vs-Rest (OvR) strategy to generate multiclass ROC curves, with Type I designated as the positive class. LC_X_DistMean achieved an AUC of 0.811 (95% CI: 0.744–0.870, sensitivity 64.1%, specificity 88.41%). For IOP sensitivity (Figure 7), ONH-related parameters exhibited superior discrimination (AUCs ≥0.96), with strain_max_ONH reaching an AUC of 0.993 (95% CI: 0.980–1.000). Strain_max_Opticnerve also performed strongly (AUC = 0.991). Additionally, HSR_RM_0.04 and LC_X_DistMean demonstrated substantial discriminative ability (AUCs of 0.883 and 0.832). More detailed ROC analysis results are provided in Table 5 (Type I and II vs. others). For Types I and II vs. others, HSR_RM_0.04 separated the groups with an AUC of 0.831 after reporting the stronger-separation direction (raw AUC <0.5), consistent with lower macular high-strain exposure in macular-involving subtypes.

3.4 Effect of staphyloma depth-radius morphology

After excluding PSR surgical models, we analyzed the distribution of HSR_RM_0.04 changes (from 10 mmHg to 21 mmHg) with depth-radius variations across PSS subtypes and ALs. Results (Figure 8) showed a diagonal gradient pattern for Type II, III and V.

Due to non-normal distributions and small samples, Spearman's ρ with 10,000-iteration permutation tests (p_perm) and adjusted R^2 were calculated to evaluate the synergistic effects of radius (R) and depth (D) across all parameters. Table 6 lists Subtype and AL subgroups with adjusted $R^2 \geq 0.75$ and significant Spearman correlation with R or D (p_perm <0.05). Radius (R) served as the primary determinant (ρ = 0.74–0.95 for LC_X_DistMean and 0.79–0.95 for HSR_RM_0.04; Y–R was significant and positive in four of six LC_Y_DistMean subgroups (ρ = 0.84–0.95). Multivariate modeling confirmed the "Depth-Radius" profile as associated with macular strain (adjusted R^2: 0.75–0.99), and this synergistic effect is subtype-specific.

TABLE 5 ROC analysis results of key biomechanical parameters for different experimental groups.

Experiment	Feature	AUC (95% CI)	Sensitivity	Specificity
IOP10 vs. IOP21	HSR_RM_0.04	0.883 (0.846–0.915)	0.7353	0.8765
	LC_X_DistMean	0.832 (0.784–0.875)	0.9	0.7412
	LC_Y_DistMean	0.728 (0.674–0.780)	0.6706	0.6941
	ISR_RM_0.25	0.525 (0.466–0.581)	0.3647	0.7235
Type I vs. others	HSR_RM_0.04	0.824 (0.768–0.874)	0.7188	0.8116
	LC_X_DistMean	0.811 (0.744–0.870)	0.6406	0.8841
	LC_Y_DistMean	0.792 (0.723–0.853)	0.8281	0.6993
	ISR_RM_0.25	0.500 (0.444–0.562)	0.9688	0.2971
Type I and II vs. others	HSR_RM_0.04	0.831 (0.786–0.872)	0.6119	0.9223
	LC_Y_DistMean	0.595 (0.531–0.659)	0.4851	0.7184
	ISR_RM_0.25	0.590 (0.528–0.649)	0.9254	0.4854
	LC_X_DistMean	0.587 (0.522–0.650)	0.5224	0.665

When raw AUC, was <0.5 (higher-score-positive scoring), the score was reversed and AUC, 1 − raw AUC., Sensitivity and specificity are at the Youden point; AUCs, describe separation of simulated conditions, not clinical diagnosis.

3.5 Postoperative effects of posterior scleral reinforcement

To quantitatively evaluate the improvement effects of PSR on the ocular biomechanical environment, we used surgical status as the grouping variable. Results revealed that only HSR_RM_0.04 was lower in the PSR models (from 19.45% to 3.58%), indicating reduced macular high-strain exposure. These averages pool different geometries; the geometry-matched comparison is the strip-modulus series in Figure 9.

Effect of elastic modulus change (3–50 MPa) on mechanical responses across five PSS subtypes showed that strip with higher elastic modulus enhanced surgical improvement (Figure 9). HSR_RM_0.04 of all PSR surgical groups reduced by 23.5%–87.7% versus non-surgical models. A monotonic decline was observed with increasing elastic modulus: Type I (from 1.24% to 1.06%), Type II (from 1.73% to 1.37%), Type III (from 13.37% to 6.47%), Type IV (from 3.42% to 2.55%), and Type V (from 6.35% to 4.57%). These results showed elastic modulus of strip as a key regulator of PSR outcome. A moderately increase of elastic modulus strengthens posterior scleral support, more effectively limiting strain accumulation in the ONH and macular region during IOP fluctuations.

4 Discussion

In this study, we developed a series of whole-eye finite element models encompassing five PSS subtypes, three axial lengths (ALs), and two intraocular pressure (IOP) levels to elucidate the biomechanical determinants governing posterior pole responses. Although morphological heterogeneity among PSS subtypes is well documented—with Curtin's classification identifying five distinct subtypes and accumulating evidence revealing their heterogeneous biomechanical impact and fundus damage patterns across different clinical prognoses—the underlying biomechanical responses that drive differential disease progression remain unclear. We hypothesized that distinct mechanical microenvironments drive differential disease progression of PSS, challenging the conventional reliance on AL.

Using multi-parameter finite element models of all five primary subtypes, we aimed to link PSS morphology to biomechanical response of LC and macular. Model parameters (Table 3) were derived from established clinical and anatomical benchmarks. PSS subtypes were classified according to Curtin's system (Curtin, 1977), with Type I being the most prevalent (~76%) (Ohno-Matsui, 2014). AL was set at 26 mm, 29 mm, and 36 mm to represent mild, moderate, and extreme elongation, aligning with the median AL (~29 mm) reported in PM (Curtin, 1977; Knaus et al., 2021; Luo et al., 2025) and the sharp increase in staphyloma incidence from 1.4% to 71.4% as AL exceeds 33.5 mm (Curtin and Karlin, 1971).

Our simulations revealed that PSS subtype-specific geometries showed the largest effect within the tested parameter ranges, surpassing AL and staphyloma morphology (Depth-Radius profile). Furthermore, we quantified the protective effect of posterior scleral reinforcement (PSR), identifying strip modulus as a critical modulator in surgical efficacy.

Our findings agree with existing literature, validating the model's plausibility. The LC_X_DistMean (12.21 ± 9.26 μm) fell within the reported range for IOP-induced deformation (12.9–53.6 μm) (Sigal et al., 2011). Our LC first principal strain (5.37% ± 2.78%) fit with 1%–5% (Norman et al., 2011), LC equivalent strain (7.05% ± 3.74%) approximated prior estimates (2.65%–3.83%) (Li et al., 2026) and remained below 30% (Islam et al., 2024), maximum LC stress (20.29 kPa) was lower than previous reports (75.70 kPa (Karimi et al., 2022); 180 kPa (Shin et al., 2017)). Maximum ONH stress (3.07 ± 1.29 kPa) and strain (11.08% ± 4.80%) were consistent with documented ranges (20–30 kPa (Karimi et al., 2022); 10%–30% (Shin et al., 2017)). Parameters we selected also demonstrated strong discriminative

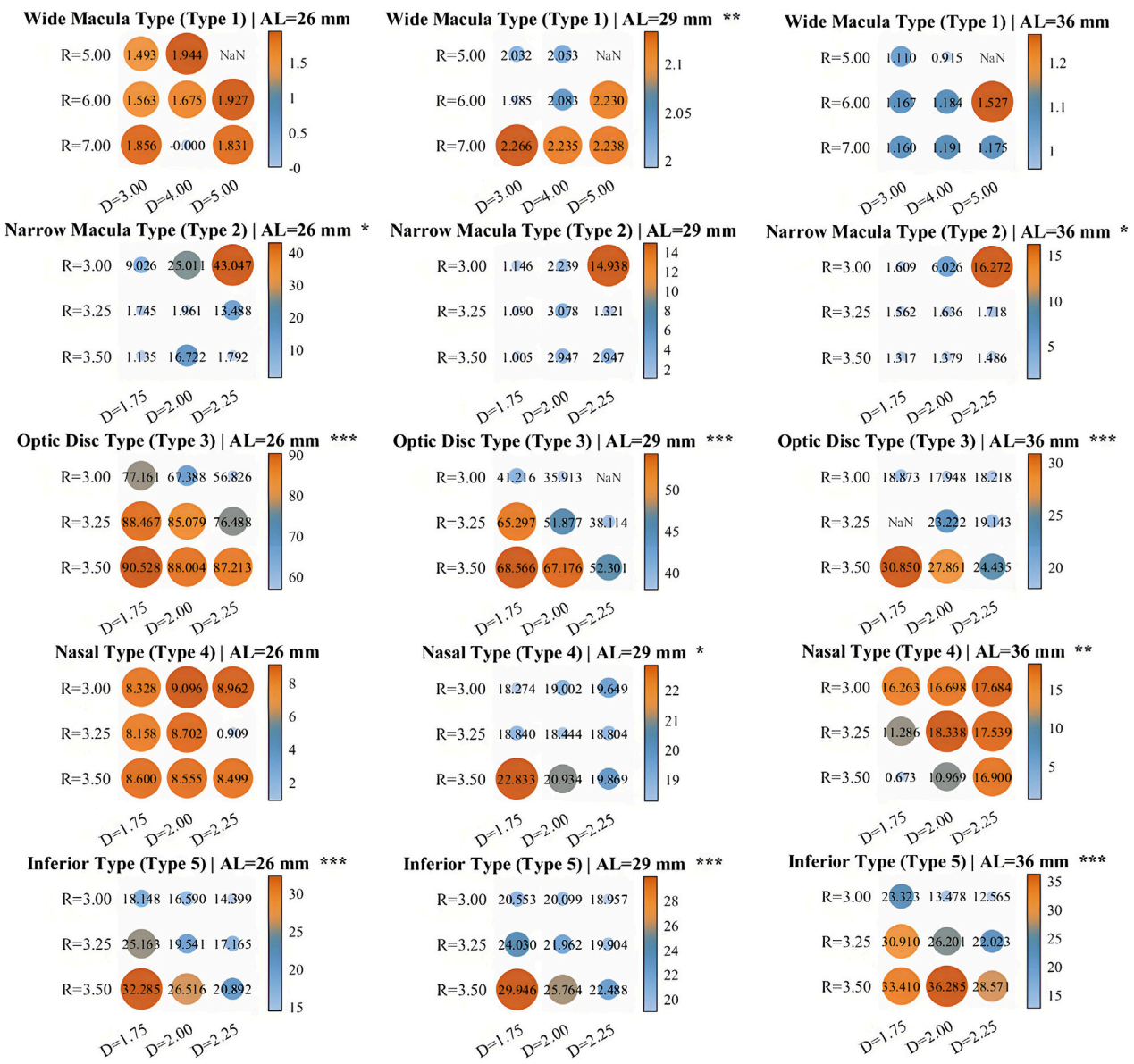


FIGURE 8
Synergistic Effects of Staphyloma Depth (D) and Radius (R) on Macular Strain (HSR_RM_0.04). The distribution of HSR_RM_0.04 (under two IOP conditions: 10 and 21 mmHg) across PSS subtypes (rows) and axial lengths (columns) in non-surgical eyes is visualized. In the bubble chart, both color and size represent the magnitude of HSR. Note: Asterisks next to the panel titles show the adjusted R^2 of Y~R + D: * ≥ 0.25, ** ≥ 0.50, and *** ≥ 0.75.

power, HSR_RM_0.04 achieved an AUC of 0.883 for IOP discriminating, and HSR_RM_0.04 and LC_X_DistMean reached AUCs of 0.824 and 0.811 for Type I vs. others for distinguishing Type I staphyloma, indicating separation within the simulated parameter space.

Elevated IOP creates mechanical strain at the ONH that impairs axonal function, leading to local extracellular matrix remodeling and retinal ganglion cell death which are hallmarks of glaucoma and account for glaucomatous vision loss (Pitha et al., 2024). The ONH, particularly the LC and prelaminar neural tissue, is directly and continuously loaded by IOP, and this influence has been demonstrated quantitatively *in vivo* clinical studies. Using spectral-domain optical coherence tomography (OCT) combined with digital volume correlation (DVC), Hannay et al. (2024) measured LC strains in 23 eyes of 15 glaucoma patients 1 week after medication-induced IOP lowering. In the subgroup achieving ≥3 mmHg IOP reduction, the LC exhibited significant tensile anterior–posterior strain (Ezz = 1.0% ± 1.1%, p = 0.003), compressive radial strain (Err = −0.3% ± 0.5%, p = 0.012), maximum in-plane principal strain (Emax = 1.7% ± 1.0%), and maximum shear strain (Γmax = 1.4% ± 0.7%). In brain-dead organ donors, Girkin et al. (Girkin et al., 2023) showed that acute IOP elevation from 10 to 50 mmHg caused peripapillary RNFL thinning at a rate of −0.0717 μm/mmHg ($p < 0.001$) and choroidal thinning of −0.5222 μm/mmHg ($p < 0.001$), directly quantifying IOP-driven structural compression. In finite element models of enucleated human eyes pressurized to 30 mmHg, varying peripapillary scleral thickness over the measured range changed

TABLE 6 Non-parametric correlation and multivariate regression analysis of morphological parameters (D, R) across PSS subtypes.

Parameter Y	PSS subtype	AL (mm)	N	Y–R (Spearman)	p_perm (R)	Y–D (Spearman)	p_perm (D)	Adjusted R^2
HSR_RM_0.04	III	26	9	0.791	0.021*	−0.58	0.121	0.823
	III	29	8	0.806	0.022*	−0.391	0.31	0.889
	III	36	8	0.945	0.005**	−0.239	0.577	0.9
	V	26	9	0.843	0.009**	−0.527	0.168	0.909
	V	29	9	0.791	0.023*	−0.58	0.128	0.851
	V	36	9	0.843	0.010**	−0.474	0.21	0.884
LC_X_DistMean	II	36	9	0.949	0.001**	−0.264	0.531	0.897
	III	26	9	0.843	0.010**	−0.527	0.164	0.989
	IV	36	9	0.791	0.021*	−0.58	0.131	0.918
	V	26	9	0.896	0.003**	−0.422	0.28	0.821
	V	36	9	0.738	0.035*	−0.632	0.086	0.882
LC_Y_DistMean	I	29	8	0.869	0.012*	−0.252	0.549	0.992
	II	36	9	0.949	0.002**	−0.264	0.525	0.941
	III	26	9	0.843	0.011*	−0.527	0.167	0.988
	III	36	8	−0.567	0.179	0.743	0.041*	0.751
	IV	26	9	−0.527	0.158	0.791	0.024*	0.803
	IV	29	9	0.843	0.012*	−0.474	0.212	0.906

ρ denotes the Spearman rank correlation coefficient. P_perm values were derived from 10,000-iteration permutation tests to empirically estimate significance levels, accounting for non-normal distributions and small sample sizes (*p_perm<0.05, **p_perm<0.01). Adjusted R^2 values were obtained from the multivariate linear regression model Y~R + D.

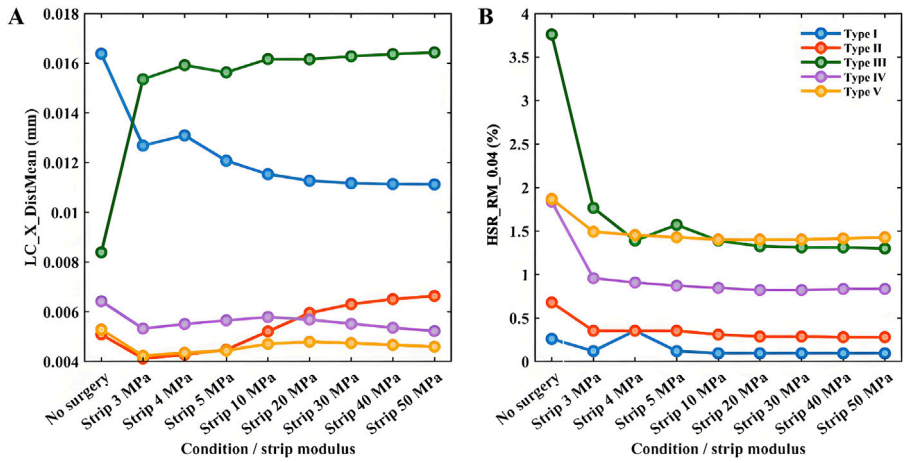


FIGURE 9
Effects of post-PSR strip modulus on **(A)** LC_X_DistMean and **(B)** HSR_RM_0.04. The first points in each panel indicate non-surgical models; subsequent points indicate post-PSR strip moduli (3-50 MPa). Lines represent PSS subtypes I-V.

the 95th-percentile first principal strain in the LC by an amount equivalent to a 15 mmHg change in IOP (Norman et al., 2011). The increase in maximum optic-nerve strain from 10 to 21 mmHg in our models is in the same direction. These comparisons describe deformation, not clinical optic-nerve function. It is found in our study, within Type I, mean LC displacements increased from 15.23 ± 9.33 μm to 28.10 ± 13.40 μm (LC_X_DistMean) and from 13.50 ± 9.92 μm to 24.36 ± 14.83 μm (LC_Y_DistMean) as IOP rose from 10 to 21 mmHg, remaining within the range of previously reported IOP-driven LC motions, while maximum optic-nerve strain rose from 4.77% ± 1.29%–10.90% ± 2.42% (Supplementary Figure S1). The mean maximal LC principal strain in Type I (4.20% ± 2.08%)

was also comparable in magnitude to the results of Hannay et al. (Hannay et al., 2024) and earlier human FE estimates of mean LC strain (~4%–5.5%) under higher IOP loads (Sigal et al., 2004). The maximum optic-nerve strain was strongly correlated with the maximum equivalent strain of the lamina cribrosa (Figure 5). When IOP increased from 10 to 21 mmHg, maximum optic-nerve strain increased (Supplementary Figure S1). Elevated IOP induces deformation of the lamina cribrosa, which compresses the optic nerve fibers passing through the laminar pores, thereby leading to optic nerve damage.

Traditional assessments of PM relying primarily on AL have been suggested to be insufficient, as they may overlook the independent role of PSS subtypes in determining eye shape and associated complications (Moriyama et al., 2011). Our results suggested that PSS subtype specific geometries showed the largest effect within the tested parameter ranges ($\varepsilon^2 > 0.33$), whose explanatory power exceeds that of AL, staphyloma depth D, radius R, and surgical status. Type I PSS exhibits higher LC strain accumulation risk (largest LC deformation: LC_X_DistMean, 0.0215 ± 0.0133 mm; LC_Y_DistMean, 0.0188 ± 0.0138 mm) than other subtypes. Our data provide the direct association between PSS morphological subtypes and quantified mechanical metrics (e.g., LC displacement, macular high-strain ratio), revealing distinct "mechanical risk profiles" for each subtype and thereby explaining why patients with the same axial length may face substantially different risks of blindness.

The Depth–Radius combination was associated with macular strain (HSR_RM_0.04) in a subtype-specific manner (Figure 8). D and R were used as paired morphological descriptors to map responses across the D–R design space. Spearman correlations and Y ~ R + D regressions varied across PSS subtypes, with the strongest fits in Types III and V. These findings suggest that morphology may inform *in silico* stratification and computational PSR planning; post-operative validation is still required. Although whether the necessity of surgery and prognosis are influenced by subtype is a clinical concern (Chen et al., 2020; Zhu et al., 2016; Ruiz-Medrano et al., 2024), the present experimental results did not provide effective support for this association. Nevertheless, the distinct biomechanical responses uncovered here provide a mechanistic rationale for optimizing surgical approaches.

More importantly, we provided a mechanical comparison of PSR strip moduli in these models. PSR significantly reduced HSR_RM_0.04 (from 19.45% to 3.58%; $p < 0.001$). However, PSR did not significantly alter LC displacement (LC_X_DistMean and LC_Y_DistMean; both $p > 0.05$). Strip modulus (ranging from 3 to 50 MPa) further attenuated IOP-induced macular high-strain exposure (HSR_RM_0.04), with reductions of 23.5%–87.7% relative to non-surgical models and a monotonic decline as modulus increased across PSS subtypes (Figure 9). These results suggest that moderately increasing posterior scleral strip rigidity can limit macular high-strain responses during IOP fluctuations; whether PSR confers protection against optic nerve injury remains unclear.

Ruiz-Moreno et al. (Ruiz-Moreno et al., 2023; Ruiz-Medrano et al., 2024) found that macular-involved PSS is associated with worse clinical outcomes and a higher risk of advanced myopic maculopathy. The compound subtype, often accompanied by macular involvement, showed particularly strong associations with disease progression. Our results also revealed differences in LC_X_DistMean between Subtype I and other subtypes (0.0215 ± 0.0133 mm vs. 0.0101 ± 0.0064 mm), as well as between combined Subtypes I and II and others (0.0145 ± 0.0116 mm vs. 0.0107 ± 0.0069 mm). ROC analysis further showed that Types I and II could be separated from the other subtypes by HSR_RM_0.04 (AUC = 0.831 after direction-corrected reporting), indicating a distinct macular strain pattern in macular-involving geometries. Although our study did not assess PM outcomes directly and did not perform pairwise inferential tests for these grouped comparisons, the morphological parallels—specifically the maximum deformation localized at the staphyloma rim observed in the vast majority of finite element models—support the concept that stress concentrates at this transition zone, consistent with histomorphometric evidence of abrupt scleral thinning and Bruch's membrane defects at the staphyloma margin (Curtin, 1977; Jonas et al., 2020; Ohno-Matsui, 2014).

Given that our simulations identified HSR_RM_0.04 as a sensitive indicator of mechanical risk, the clinical implications are particularly relevant in the context of elevated intraocular pressure (IOP). The incidence of high IOP in high myopia, especially when complicated with glaucoma, is significantly higher than that in the emmetropic population (Du et al., 2024), and it is also an independent risk factor for the progression of atrophic macular lesions (Zhao et al., 2026). IOP is a typical mechanical influencing factor, and high myopia is a risk factor for early IOP spikes after multiple intraocular surgeries, including phacoemulsification and vitrectomy with silicone oil injection (Du et al., 2024). In our model, the HSR_RM_0.04 parameter is sensitive to IOP changes (Table 5), which can quantitatively characterize the stress state in the macular region, and thus deserves more attention in future studies related to high myopia.

There are several limitations in our paper. Several assumptions were made in our analysis. First, we simplified the ONH geometry and LC thickness was fixed at 100 μm in every model in this study. The thickness of LC in highly myopic eyes was 137.07 ± 18.51 μm without open-angle glaucoma and 107.76 ± 9.86 μm with open-angle glaucoma (Chen et al., 2023). Jonas et al. (Jonas et al., 2004) likewise reported a thinner lamina and a shorter distance between the intraocular space and the cerebrospinal fluid space in highly myopic eyes. Beyond thickness, three-dimensional OCT in the Beijing Eye Study showed a more negative horizontal LC tilt in highly myopic eyes (−4.38° ± 5.94° versus −0.04° ± 5.86°; $P < 0.001$), while LC depth did not differ significantly (Hannay et al., 2024). We have added the influence of ONH in limits and future directions as follows. In future research, we will develop a finite element model of the ONH that better reflects the real anatomical structure, e.g., by constructing an OCT-based ONH model or by accounting for variation in LC thickness. Second, the sclera is the primary load-bearing tissue of the eye, and its mechanical behavior is governed jointly by collagen fiber orientation and nonlinearity. A review by Boote et al. systematically elucidated the relationship between the hierarchical structure of scleral collagen and its biomechanics (Boote et al., 2020). Recent studies have further revealed a counterintuitive inconsistency between mechanical anisotropy and collagen fiber preferred orientation in the equatorial sclera, as well as the potential of direct fiber models for characterizing scleral anisotropy (Li et al., 2025). We modeled the sclera as a homogeneous, linear, and isotropic material in this study. The findings are based on these material assumptions and do not capture scleral anisotropy or the

IOP-dependent nonlinear behavior of collagen fibers. In our future work, the nonlinearity and anisotropy of the sclera will be taken into account. Finally, the number of FEM models in non-surgical groups was limited (N = 8–9 per subtype/AL), and we will establish finite element models of the PSR by considering more parameters.

By integrating five PSS subtypes, three ALs, and two IOP levels into whole-eye finite element models, we systematically delineated the biomechanical landscape of PM. The research found that the PSS subtype and staphyloma depth-radius morphological combinations influenced biomechanical responses of posterior pole structures. PSR simulation revealed a significant reduction in macular high-strain exposure following IOP elevation with the strip configuration used in this study. These insights provide a biomechanical account of subtype-related differences at the posterior pole and a framework for comparing PSR strip settings.

Data availability statement

The original contributions presented in the study are included in the article/Supplementary Material, further inquiries can be directed to the corresponding authors.

Author contributions

YW: Software, Visualization, Investigation, Writing – original draft, Formal Analysis. XL: Writing – review and editing, Investigation. JR: Writing – review and editing. LQ: Writing – review and editing, Investigation. XQ: Writing – review and editing, Methodology, Conceptualization, Resources, Funding acquisition.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Supplementary material

The Supplementary Material for this article can be found online at: https://www.frontiersin.org/articles/10.3389/fbioe.2026.1948781/full#supplementary-material

References

Boote, C., Sigal, I. A., Grytz, R., Hua, Y., Nguyen, T. D., and Girard, M. J. A. (2020). Scleral structure and biomechanics. *Prog. Retin Eye Res.* 74, 100773. doi: 10.1016/j.preteyeres.2019.100773

Braeu, F. A., Avril, S., and Girard, M. J. A. (2024). 3D growth and remodeling theory supports the hypothesis of staphyloma formation from local scleral weakening under normal intraocular pressure. *Biomech. Model. Mechanobiol.* 23, 2137–2154. doi: 10.1007/s10237-024-01885-9

Chen, C. A., Lin, P. Y., and Wu, P. C. (2020). Treatment effect of posterior scleral reinforcement on controlling myopia progression: a systematic review and meta-analysis. *PLoS One* 15, e0233564. doi: 10.1371/journal.pone.0233564

Chen, Y., Mi, B., Li, H., Du, B., Liu, L., Xing, X., et al. (2023). Thinning of the Lamina cribrosa and deep layer microvascular dropout in patients with open angle glaucoma and high myopia. *J. Glaucoma* 32, 585–592. doi: 10.1097/IJG.0000000000002190

Cui, D., Hou, X., Li, J., Qu, X., Yu, T., and Song, A. (2021). Relationship between peripapillary choroidal thickness and retinal nerve fiber layer in young people with myopia. *J. Int. Med. Res.* 49, 3000605211032780. doi: 10.1177/03000605211032780

Curtin, B. J. (1977). The posterior staphyloma of pathologic myopia. *Trans. Am. Ophthalmol. Soc.* 75, 67–86.

Curtin, B. J., and Karlin, D. B. (1971). Axial length measurements and fundus changes of the myopic eye. *Am. J. Ophthalmol.* 71, 42–53. doi: 10.1016/0002-9394(71)91092-0

Delaunay, K., Picard, E., Lassiaz, P., Jonet, L., Cannaya, V., Ruiz-Moreno, J. M., et al. (2025). LRP2 is a potential molecular target for nonsyndromic pathological myopia. *JCI Insight* 10, e192929. doi: 10.1172/jci.insight.192929

Du, Y., Meng, J., He, W., Qi, J., Lu, Y., and Zhu, X. (2024). Complications of high myopia: an update from clinical manifestations to underlying mechanisms. *Adv. Ophthalmol. Pract. Res.* 4, 156–163. doi: 10.1016/j.aopr.2024.06.003

Ehongo, A. (2023). Understanding posterior staphyloma in pathologic myopia: current overview, new input, and perspectives. *Clin. Ophthalmol.* 17, 3825–3853. doi: 10.2147/OPTH.S399278

Fiel Peres, F. (2026). Effect sizes for nonparametric tests. *Biochem. Med.* 36, 010101–010116. doi: 10.11613/BM.2026.010101

Flores-Moreno, I., Puertas, M., Ruiz-Medrano, J., Almazán-Alonso, E., García-Zamora, M., and Ruiz-Moreno, J. M. (2024). Influence of posterior staphyloma in myopic maculopathy and visual prognosis. *Eye* 38, 145–152. doi: 10.1038/s41433-023-02648-z

Friberg, T. R., and Lace, J. W. (1988). A comparison of the elastic properties of human choroid and sclera. *Exp. Eye Res.* 47, 429–436. doi: 10.1016/0014-4835(88)90053-x

Fricke, T. R., Jong, M., Naidoo, K. S., Sankaridurg, P., Naduvilath, T. J., Ho, S. M., et al. (2018). Global prevalence of visual impairment associated with myopic macular degeneration and temporal trends from 2000 through 2050: systematic review, meta-analysis and modelling. *Br. J. Ophthalmol.* 102, 855–862. doi: 10.1136/bjophthalmol-2017-311266

Girkin, C. A., Garner, M. A., Gardiner, S. K., Clark, M. E., Hubbard, M., Karuppanan, U., et al. (2023). Displacement of the lamina cribrosa with acute intraocular pressure increase in brain-dead organ donors. *Invest. Ophthalmol. Vis. Sci.* 64, 19. doi: 10.1167/iovs.64.15.19

Han, Y. X., Wang, X., Xue, C. C., Jonas, J. B., and Wang, Y. X. (2024a). Lamina cribrosa configurations in highly myopic and non-highly myopic eyes: the beijing eye study. *Invest. Ophthalmol. Vis. Sci.* 65, 28. doi: 10.1167/iovs.65.8.28

Han, Y. X., Guo, X. X., Wang, Y. X., Jonas, J. B., Chen, X., and Wang, X. F. (2024b). Automated posterior scleral topography assessment for enhanced staphyloma visualization and quantification with improved maculopathy correlation. *Transl. Vis. Sci. Technol.* 13, 41. doi: 10.1167/tvst.13.10.41

Hannay, V., Czerpak, C. A., Quigley, H. A., and Nguyen, T. D. (2024). A noninvasive clinical method to measure *in vivo* mechanical strains of the Lamina cribrosa by OCT. Ophthalmol. *Sci* 4, 100473. doi: 10.1016/j.xops.2024.100473

Hayashi, M., Ito, Y., Takahashi, A., Kawano, K., and Terasaki, H. (2013). Scleral thickness in highly myopic eyes measured by enhanced depth imaging optical coherence tomography. *Eye* 27, 410–417. doi: 10.1038/eye.2012.289

Holden, B. A., Fricke, T. R., Wilson, D. A., Jong, M., Naidoo, K. S., Sankaridurg, P., et al. (2016). Global prevalence of myopia and high myopia and temporal trends from 2000 through 2050. *Ophthalmology* 123, 1036–1042. doi: 10.1016/j.ophtha.2016.01.006

Islam, M. R., Ji, F., Bansal, M., Hua, Y., and Sigal, I. A. (2024). Fibrous finite element modeling of the optic nerve head region. *Acta Biomater.* 175, 123–137. doi: 10.1016/j.actbio.2023.12.034

Jonas, J. B., Berenshtein, E., and Holbach, L. (2004). Lamina cribrosa thickness and spatial relationships between intraocular space and cerebrospinal fluid space in highly myopic eyes. *Invest. Ophthalmol. Vis. Sci.* 45, 2660–2665. doi: 10.1167/iovs.03-1363

Jonas, J. B., Ohno-Matsui, K., Holbach, L., and Panda-Jonas, S. (2020). Histology of myopic posterior scleral staphylomas. *Acta Ophthalmol.* 98, e856–e863. doi: 10.1111/aos.14405

Karimi, A., Rahmati, S. M., Razaghi, R., Girkin, C. A., and Crawford Downs, J. (2022). Finite element modeling of the complex anisotropic mechanical behavior of the human sclera and pia mater. *Comput. Methods Programs Biomed.* 215, 106618. doi: 10.1016/j.cmpb.2022.106618

Knaus, K. R., Hipsley, A., and Blemker, S. S. (2021). The action of ciliary muscle contraction on accommodation of the lens explored with a 3D model. *Biomech. Model. Mechanobiol.* 20, 879–894. doi: 10.1007/s10237-021-01417-9

Kolb, H., Nelson, R. F., Ahnelt, P. K., Ortuño-Lizarán, I., and Cuenca, N. (2020). "The architecture of the human fovea," in **Webvision: The Organization of the Retina and Visual System** (Salt Lake City, UT: University of Utah Health).

Krag, S., and Andreassen, T. T. (2003). Mechanical properties of the human lens capsule. *Prog. Retin. Eye Res.* 22, 749–767. doi: 10.1016/s1350-9462(03)00063-6

Kruskal, W. H., and Wallis, W. A. (1952). Use of ranks in one-criterion variance analysis. *J. Am. Stat. Assoc.* 47, 583–621. doi: 10.1080/01621459.1952.10483441

Li, R., Hua, Y., Zhang, F., Qian, X., Gong, C., Wan, X., et al. (2025). Multiscale analysis of equatorial sclera anisotropy: revealing discrepancies in fiber orientation and mechanical properties. *Sci. Adv.* 11, eadp8631. doi: 10.1126/sciadv.adp8631

Li, Q., Zhan, B., Liu, T., Han, Y., Xin, S., Chen, Z., et al. (2026). An automated optical coherence tomography to finite element analysis pipeline reveals key morphological determinants of optic nerve head biomechanics in glaucoma. *Eye* 40, 238–244. doi: 10.1038/s41433-025-04146-w

Liu, X., Wang, L., Wang, C., Sun, G., Liu, S., and Fan, Y. (2013). Mechanism of traumatic retinal detachment in blunt impact: a finite element study. *J. Biomech.* 46, 1321–1327. doi: 10.1016/j.jbiomech.2013.02.006

Liu, F., Wang, Y., Niu, L., Zhao, J., Xue, K., and Zhou, X. (2025). Characteristics of retina, choroid, and retrobulbar blood flow in myopes with posterior staphyloma using ultra-widefield OCTA and CDI. *Invest. Ophthalmol. Vis. Sci.* 66, 21. doi: 10.1167/iovs.66.3.21

Liu, T., Wang, Y. X., Jonas, J. B., Hoang, Q. V., Girard, M. J. A., and Wang, X. (2025a). Gaze-induced optic nerve head deformations are greater in high myopia and strains increase with axial length. *Invest. Ophthalmol. Vis. Sci.* 66, 21. doi: 10.1167/iovs.66.9.21

Liu, T., Wang, K., Wang, Y. X., Liu, X., Han, J., Girard, M. J. A., et al. (2025b). Ciliary muscle traction during accommodation is able to induce optic nerve head deformation. *Eye* 39, 1175–1182. doi: 10.1038/s41433-024-03569-1

Luo, N., Wang, Y., Alimu, S., Zhao, L., Huang, Y., Guo, Z., et al. (2023). Assessment of ocular deformation in pathologic myopia using 3-dimensional magnetic resonance imaging. *JAMA Ophthalmol.* 141, 768–774. doi: 10.1001/jamaophthalmol.2023.2869

Luo, N., Liu, B., Wang, L., Yang, Z., Wang, P., Ho, M., et al. (2025). Fundus curvature as a clinical marker for posterior staphyloma based on widefield optical coherence tomography. *Am. J. Ophthalmol.* 271, 478–487. doi: 10.1016/j.ajo.2024.12.020

Ma, J., Wu, F., Liu, Z., Fang, Y., Chu, X., Zheng, L., et al. (2022). Biomechanical considerations of patching material for posterior scleral reinforcement surgery. *Front. Med.* 9, 888542. doi: 10.3389/fmed.2022.888542

Miao, Z., Li, L., Meng, X., Guo, L., Cao, D., Jia, Y., et al. (2019). Modified posterior scleral reinforcement as a treatment for high myopia in children and its therapeutic effect. *Biomed. Res. Int.* 2019, 5185780. doi: 10.1155/2019/5185780

Miller, K. (1999). Constitutive model of brain tissue suitable for finite element analysis of surgical procedures. *J. Biomech.* 32, 531–537. doi: 10.1016/s0021-9290(99)00010-x

Moriyama, M., Ohno-Matsui, K., Hayashi, K., Shimada, N., Yoshida, T., Tokoro, T., et al. (2011). Topographic analyses of shape of eyes with pathologic myopia by high-resolution three-dimensional magnetic resonance imaging. *Ophthalmology* 118, 1626–1637. doi: 10.1016/j.ophtha.2011.01.018

Norman, R. E., Flanagan, J. G., Sigal, I. A., Rausch, S. M., Tertinegg, I., and Ethier, C. R. (2011). Finite element modeling of the human sclera: influence on optic nerve head biomechanics and connections with glaucoma. *Exp. Eye Res.* 93, 4–12. doi: 10.1016/j.exer.2010.09.014

Ohno-Matsui, K. (2014). Proposed classification of posterior staphylomas based on analyses of eye shape by three-dimensional magnetic resonance imaging and wide-field fundus imaging. *Ophthalmology* 121, 1798–1809. doi: 10.1016/j.ophtha.2014.03.035

Ohno-Matsui, K., and Jonas, J. B. (2019). Posterior staphyloma in pathologic myopia. *Prog. Retin. Eye Res.* 70, 99–109. doi: 10.1016/j.preteyeres.2018.12.001

Ohno-Matsui, K., Lai, T. Y., Lai, C. C., and Cheung, C. M. (2016). Updates of pathologic myopia. *Prog. Retin. Eye Res.* 52, 156–187. doi: 10.1016/j.preteyeres.2015.12.001

Pitha, I., Du, L., Nguyen, T. D., and Quigley, H. A. (2024). IOP and glaucoma damage: the essential role of optic nerve head and retinal mechanosensors. *Prog. Retin. Eye Res.* 99, 101232. doi: 10.1016/j.preteyeres.2023.101232

Polyak, S. L. (1941). "The retina: the anatomy and the histology of the retina in man, ape, and monkey, including the consideration of visual functions," in *The History of Physiological Optics, and the Histological Laboratory Technique*. Chicago: University of Chicago Press.

Ruiz-Medrano, J., Puertas, M., Flores-Moreno, I., Almazán-Alonso, E., García-Zamora, M., Kudsieh, B., et al. (2024). The importance of the type of posterior staphyloma in the development of myopic maculopathy. *Diagnostics* 14, 1581. doi: 10.3390/diagnostics14151581

Ruiz-Moreno, J. M., Puertas, M., Flores-Moreno, I., Almazán-Alonso, E., García-Zamora, M., and Ruiz-Medrano, J. (2023). Posterior staphyloma as determining factor for myopic maculopathy. *Am. J. Ophthalmol.* 252, 9–16. doi: 10.1016/j.ajo.2023.02.017

Shin, A., Yoo, L., Park, J., and Demer, J. L. (2017). Finite element biomechanics of optic nerve sheath traction in adduction. *J. Biomech. Eng.* 139, 1010101. doi: 10.1115/1.4037562

Shinohara, K., Shimada, N., Moriyama, M., Yoshida, T., Jonas, J. B., Yoshimura, N., et al. (2017). Posterior staphylomas in pathologic myopia imaged by widefield optical coherence tomography. *Investigative Ophthalmology and Visual Science* 58 (9), 3750–3758. doi: 10.1167/iovs.17-22319

Sigal, I. A., Flanagan, J. G., Tertinegg, I., and Ethier, C. R. (2004). Finite element modeling of optic nerve head biomechanics. *Invest. Ophthalmol. Vis. Sci.* 45, 4378–4387. doi: 10.1167/iovs.04-0133

Sigal, I. A., Flanagan, J. G., Tertinegg, I., and Ethier, C. R. (2005). Reconstruction of human optic nerve heads for finite element modeling. *Technol. Health Care* 13, 313–329. doi: 10.3233/THC-2005-13410

Sigal, I. A., Yang, H., Roberts, M. D., Burgoyne, C. F., and Downs, J. C. (2011). IOP-Induced Lamina cribrosa displacement and scleral canal expansion: an analysis of factor interactions using parameterized eye-specific models. *Invest. Ophthalmol. Vis. Sci.* 52, 1896–1907. doi: 10.1167/iovs.10-5500

Spaide, R. F. (2014). "Staphyloma: part 1," in *Pathologic Myopia*. Editors R. Spaide, K. Ohno-Matsui, and L. Yannuzzi (New York, NY: Springer). doi: 10.1007/978-1-4614-8338-0_12

Steidl, S. M., and Pruett, R. C. (1997). Macular complications associated with posterior staphyloma. *Am. J. Ophthalmol.* 123, 181–187. doi: 10.1016/s0002-9394(14)71034-7

Tanaka, N., Shinohara, K., Yokoi, T., Uramoto, K., Takahashi, H., Onishi, Y., et al. (2019). Posterior staphylomas and scleral curvature in highly myopic children and adolescents investigated by ultra-widefield optical coherence tomography. *PLoS One* 14, e0218107. doi: 10.1371/journal.pone.0218107

Wang, H., Prendiville, P. L., McDonnell, P. J., and Chang, W. V. (1996). An ultrasonic technique for the measurement of the elastic moduli of human cornea. *J. Biomech.* 29, 1633–1636. doi: 10.1016/S0021-9290(96)80017-0

Ward, B., Tarutta, E. P., and Mayer, M. J. (2009). The efficacy and safety of posterior pole buckles in the control of progressive high myopia. *Eye* 23, 2169–2174. doi: 10.1038/eye.2008.433

Wei, W. B., Xu, L., Jonas, J. B., Shao, L., Du, K. F., Wang, S., et al. (2013). Subfoveal choroidal thickness: the beijing eye study. *Ophthalmology* 120, 175–180. doi: 10.1016/j.ophtha.2012.07.048

Wei, J., Hua, Y., Yang, B., Wang, B., Schmitt, S. E., Wang, B., et al. (2022). Comparing acute IOP-Induced Lamina cribrosa deformations premortem and postmortem. *Transl. Vis. Sci. Technol.* 11, 1. doi: 10.1167/tvst.11.12.1

Xue, A., Bao, F., Zheng, L., Wang, Q., Cheng, L., and Qu, J. (2014). Posterior scleral reinforcement on progressive high myopic young patients. *Optom. Vis. Sci.* 91, 412–418. doi: 10.1097/OPX.0000000000000201

Yoon, C. K., Lee, E. K., Bae, K., and Park, U. C. (2024). Clinical features of primary and compound forms of wide macular posterior staphyloma in high myopia. *BMC Ophthalmol.* 24, 246. doi: 10.1186/s12886-024-03519-1

Zhang, L., Albon, J., Jones, H., Gouget, C. L., Ethier, C. R., Goh, J. C., et al. (2015). Collagen microstructural factors influencing optic nerve head biomechanics. *Invest. Ophthalmol. Vis. Sci.* 56, 2031–2042. doi: 10.1167/iovs.14-15734

Zhao, J., Zhao, X., Jiang, Y., Gong, Q., Xue, A., Xu, X., et al. (2026). Incidence and associated factors of myopic maculopathy progression after posterior scleral contraction in high myopia. *Ophthalmology* 133, 506–514. doi: 10.1016/j.ophtha.2025.11.002

Zhu, S. Q., Zheng, L. Y., Pan, A. P., Yu, A. Y., Wang, Q. M., and Xue, A. Q. (2016). The efficacy and safety of posterior scleral reinforcement using genipin cross-linked sclera for macular detachment and retinoschisis in highly myopic eyes. *Br. J. Ophthalmol.* 100, 1470–1475. doi: 10.1136/bjophthalmol-2015-308087